THE COST LIMIT

What a cheap 64-bit hash finalizer costs at minimum, and why.

An empirical result with a mechanism, fully reproducible. Not a mathematical proof, and this paper says exactly where the difference lies.

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What this is about

A **hash finalizer** is the last mixing step that turns a number (a customer ID, a timestamp, a pointer) into a well-shuffled value. It sits in hash tables, in spreading data over machines (sharding), in sampling. A handful of operations, executed billions of times.

The obvious question would be: which one is best? That cannot be answered with a cheap instrument — good mixers are not distinguishable that way (demonstrated elsewhere). So the question that does work:

How cheap can a GOOD finalizer be?

Cost = a weighted operation count in a coarse latency model, not a headcount of instructions: multiplication 3, 128-bit fold 4, shift/XOR 2, rotation 1, constant 1. The weights are a convention, stated so they can be argued with. Cheaper means faster, everywhere hashing happens. This paper measures that floor, and gives the reason it sits there.

There are two floors, not one.

The ladder has two heights that matter. The **entry test** asks whether a mixer survives 32 disguises at 2 GB each. The **diploma** (RRC-64-40) asks whether it survives 256 disguises at 1 TB each, a quarter of a million gigabytes. Both floors are now measured, and the distance between them is **one cost point.**

The measuring bar

The instrument is the **ladder**, not a cheap quick test; those saturate and do not separate good mixers. The finalizer is fed counting numbers in 32 disguises, read backwards, turned into their opposite, shifted by every position (Evensen's RRC scheme). An established test program from random-number research (PractRand) looks at how much data it needs before it can demonstrate a difference from real randomness.

That number does not saturate. It separates sharply:

Finalizer	Cost	Dies on the ladder
Stafford mix13	12	2^16 to 2^21 (median 2^19)
Murmur3 finalizer	12	2^14 to 2^18 (median 2^16)
mulfold x2 (bare)	8	falls on 11 of 32 disguises
constant + mulfold x2	9	holds all 32 and the depth (2^34) — tears in the diploma at 2^38
fold–xsh2–fold	10	**holds the diploma: 256/256 streams at 1 TB**

“Passes” here means: holds the 32 entry streams up to the test limit. That is the ladder's entry test; the full diploma (RRC-64-40) is a stage of its own, and both results are below.

The result

Across several systematic search runs, **no chain below cost 9 (K9)** passes the ladder. One run alone tested **9,062 structures**, plus **500 random constant variants** of the cheapest structure, plus a completed niche search over the extended operation alphabet (MAP-Elites: 467 chains, 60 generations, 34 occupied behaviour niches).

The cheapest passer is K9 — called **lu9** in this work. Below it:

Cost	Structure	Ladder
K5–K6	one multiplication / half a round	falls well before 2^16
K8	mulfold x2 (bare double fold)	falls on 11/32 — and with 500 other constants just the same
K9	constant up front + mulfold x2	passes 32/32 and 2^34 — falls in the diploma at 2^38
K10	fold — xor-shift x2 — fold	passes the diploma: 256 streams at 1 TB, no FAIL

Important and honest: the K9 structure itself is not new — it is the pattern of the wyhash family (a constant before the fold). The result of this paper is *not* a new mixer but the **measured floor** and the reason for it. The 500 constant variants additionally show: the weakness at K8 is structural, not a question of the constant.

How sharp the edge is shows in the narrowest failure of the niche search: a K9 chain of a different build (fold, byte swap, fold) holds 30 of 32 disguises — and falls on the last two. Above the limit further full passers with foreign skeletons exist (the cheapest at K10); they do not move the floor.

The mechanism: why K8 is not enough

On its own a “found nothing” is bland. Behind this one there is a reason, and the reason can be named.

A good finalizer needs two complete multiplicative rounds.

Take away one of the load-bearing operations and the mixing does not degrade a little, it degrades by a multiple (measured). The cheapest multiplicative round in this class is the 128-bit fold, cost 4. Two of them = K8.

Two rounds are the *cheap* route, not a law of nature. The niche search found exactly one chain that holds all 32 entry streams with only ONE multiplication, propped up by a scaffold of linear operations, total cost K17. So the one-round route exists, but costs almost twice the limit. Below K9 no measured route gets through.

But the bare K8 falls on one particular class of input.

Numbers of low multiplicative weight produce a thin 128-bit product: small values, or ones with many zero bits at the bottom, which is to say exactly counters and IDs. The fold then has nothing to fold. That is why mulfold x2 falls on the counting streams, and does so with every constant.

A single cheap operation up front repairs it at the cause.

An XOR with a constant (cost 1) turns every small number into a generic input before the first fold takes hold. With it the chain passes all 32 streams. That +1 is what closes the structural weakness, and that is why the floor sits at K9 and not at K8.

The targeted fix against exactly this cause erases every failure — which is the evidence that the mechanism is real and not just a story told afterwards.

The hardness test: hardware crypto against the limit

Modern CPUs execute a complete AES encryption round as ONE instruction (AES-NI). That is the natural attack on the K9 limit: a cryptographic mixing round for nominally ~5 cost points, and the AES substitution has none of the fold's weaknesses on counter inputs. If anything undercuts the limit, this does.

First: this finding already existed.

Martin Leitner-Ankerl measured two AES-based 64-bit mixers over RRC counter streams with PractRand 0.95 in his mixer benchmark (`github.com/martinus/better-faster-stronger-mixer`, README written January 2020). Counted from his tables on 30 August 2026: **aes2** dies between 2^13 and 2^16 in all 128 of his cells; **aes3** carries the note *only tested up to -tlmax 20* and does *not* fail there in 124 of 128 cells. The prior art therefore shows one folded AES construction dying early and says nothing about a deeper one — a range of 2^13 to 2^20 would read his test ceiling as a death. The measurement below stands on its own; what is new here is the *mechanism*: why they die, and that the round key has no effect once the state is folded back.

Result: it does not undercut. It dies immediately.

Every AES chain tested from K5 to K10 falls at the ladder's very first checkpoint (1–2 KB), worse than any classical mixer in this paper. The reason is nameable again: 64 bits fill only half of the 128-bit AES state, and after a single round plus folding back to 64 bits every output byte depends on exactly two input bytes — a thin web of dependencies that the test program sees instantly.

Two findings on the side that support the paper: the AES round key cancels itself out completely under the fold-back (a parameter without any effect, derived on paper and confirmed empirically). And the cheap quick test (avalanche) certified the dead two-round chain a perfect 0.4998. More evidence that cheap tests do not separate and the ladder has to judge.

Before the verdict, the AES usage itself was proven (official FIPS-197 test vector, recomputed bit for bit) and the same run showed that the measuring path still lets the known K9 passer pass. The verdict lands on the chain, not on the instrument.

For the limit this means: it holds against the multiplicative class it was won from, and against a structurally foreign hardware primitive as well. (AES hashing on a full 128-bit state with several rounds exists and works, but from ~K10 upwards, above the limit, and not on every CPU.)

Hardness test number two: a whole family of near-champions.

The K9 three-op chains over the constant pool were enumerated. The space holds 30,980 chains in three shape families; the run decided 1,309 of them, all inside the first family, before a cap stopped it. Among those, 23 chains of the form fold–byte swap–fold pass the 2 GB entry test in full, the first family on a par with the wyhash pattern, never found by any stochastic search. In the depth test (64 streams at 16 GB) **all 23** failed, between 2^31 and 2^34. Every one of the 260 individual failures lies on the streams WITHOUT complement, the signature of the known fold weakness. So the byte swap *hides* the weakness just far enough for the entry test; the constant up front (wyhash pattern) *removes* it. With that the limit holds against an enumerated fragment of one family too, which is less than it sounds, and is written here as what it is.

What that does and does not prove about position.

The obvious reading is that the repair has to act on the INPUT, not on the intermediate product. That is too strong, and meer10 is the counter-example. It carries the same skeleton as the specious family, fold–something–fold, and repairs in the middle. It passes the diploma anyway, going further than the input-side repair ever did (2^40 against 2^38). What separates them is not where the step sits but what it does for the money:

Middle step	Cost	What it does	Result
bswap	9	**rearranges** bytes, adds no diffusion	dies 2^31–2^34
x ^= (x>>28) ^ (x>>6)	10	**diffuses —** carries move down, not only up	**passes 2^40**

So the honest mechanism is about strength, not position: at one extra point the middle can only afford a permutation, and a permutation merely hides the thin product from the entry test. At two extra points it can afford real mixing, and that goes the distance. The specious family is a graveyard and a control group at once: it shows what the missing point buys.

The second bound, and what the diploma costs

It is measured now.

meer10 (fold, two xor-shifts, fold, cost 10) passes RRC-64-40. 256 streams at 1 TB each, ten at a time across at least three run segments between 22 and 28 August 2026, **not a single FAIL**, and 256 intact PractRand logs on disk to show for it.

	Entry test (2 GB x 32)	Diploma (1 TB x 256)
Floor	**9**	**10** (upper bound)
Cheapest passer	lu9, K9	meer10, K10
Below it	nothing, across ~9,000 structures, 500 constant variants, hardware AES and 1,309 enumerated chains of one family	lu9 (K9) falls at 2^38; the +const variant mfa9 (K9) falls at 2^39

Where the honesty sits in that table.

The 10 is an **upper bound, measured by construction**: a chain of cost 10 passes, so 10 suffices. It is *not* a proven lower bound. What is measured is that the two K9 chains taken to diploma distance both fell — and that of all the K9 chains that pass the entry test, not one but the wyhash pattern survived even the depth test. Whether some cost-9 chain passes the diploma is open. Anyone who writes here that nine is proven impossible is overstating what was run.

What is new, and what is not.

The diploma is Evensen's RRC scheme run to 1 TB, not a distinction invented here. NASAM (Evensen) passes it, and so does Kågström's mx3 rev2; both are markedly more expensive than 10 in this cost model. What is new is the **price**: the same distance for ten points instead of roughly fifteen. And the bracket is narrow: **the cheapest entry passer and the cheapest diploma passer have the same three-operation skeleton and differ in their middle member.** lu9 puts a constant there and tears at 2^38; meer10 puts a double shift-xor there and holds 2^40. In the weights of this paper that is one cost point; under the alternative pricing discussed in the repository README it is two. The structural statement survives either, which is why it is the one made here.

Evensen's own Moremur does not pass RRC-64-40. The diploma is a harsh bar, and the list of things that clear it is short.

What this is — and what it is NOT

It IS: empirical evidence plus mechanism.

A systematic search over thousands of structures finds nothing below K9 that passes the ladder — and a nameable mechanism explains why. Fully reproducible.

Cliff or steep curve? The counter-position.

This paper speaks of a cliff. Maiga's published numbers suggest more of a *slope*: his constructions reach 2^14, 2^36 and over 2^44 in turn — a ramp with steep sections. Anyone wanting to dispute the cliff reading has an argument there, and it belongs in the text.

What speaks for the cliff is the jump in order of magnitude: below the limit things do not die a little earlier, they die by powers of ten. In the cheapest class (cost 5) every one of the 91,000 enumerated chains has now been measured, and the median death depth is 2^12, against 2^34 and more above the limit. That is not a gradual transition. **The question is only fully settled once classes 5 to 9 have been counted through without gaps and their death depths plotted against cost**. That measurement is running and still outstanding.

The class, drawn.

Cost class 5 is small enough to draw whole. One row per skeleton, one band per chain, coloured by the volume at which PractRand first told its output from random; deepest at the top, and the number of chains printed beside each row.

The block along the bottom is the finding this paper rests on. Fourteen skeletons, 19,880 chains, a fifth of the class, die at the first checkpoint no matter which constants they are given. What they have in common is that after their last nonlinear step nothing carries high bits back down: a plain multiply moves information upward only, so without a shift-xor, a rotation or a fold to bring the top back to the bottom, the low bits keep the counter's structure. All 13 skeletons that do have that back-transport outlive the first checkpoint, without exception. The rule runs one way only, since 31 skeletons without it get past as well.

The cost axis, measured for the first time.

Every cost figure in this paper comes from a weight table, and until 31 August 2026 nothing had been measured against it: the quality axis carried 256 terabytes, the cost axis a convention. A dependent chain `x = mix(x)`, 100 million iterations, three runs on a Ryzen 7 5700X, spread under 0.17 ns:

mixer	weighted cost	ns per call	ns per point
wyrand step	5	1.59	0.32
mulfold x2	8	2.58	0.32
mfa9	9	2.82	0.31
lu9	9	2.85	0.32
mix13	12	3.01	0.25
fmix64	12	3.01	0.25
moremur	12	3.07	0.26
meer10	**10**	**3.28**	**0.33**
NASAM	~15	3.84	0.26

The measurement disagrees with the table twice, and the second one costs this paper something. The weights overprice the classical shift-multiply mixers: at cost 12 they run at 0.25 ns per point against 0.32 for the fold-based constructions. And **meer10 at cost 10 is slower than mix13, fmix64 and moremur at cost 12**, by about 0.25 ns. The cost column calls meer10 cheaper; the clock on this machine does not.

What holds is the comparison the paper actually rests on: meer10 against NASAM, 3.28 ns against 3.84, and NASAM is the mixer with a comparable published reach. Ten points instead of roughly fifteen is a real saving in latency as well as in operations. What does not hold is reading the cost column as a latency ranking across construction families. Tool and numbers: `rrc/bench_cost.cpp`, `results/cost_bench.json`.

It is NOT a proof.

The search *searched* the space, it did not *exhaust* it. “Not found in thousands of structures” is not “does not exist”. The 64-bit constant space cannot be enumerated. Anyone reading this paper as a *proof* would be wrong. It is strong evidence supported by a mechanism, and it claims nothing more.

The cost model is a named convention.

The number 9 hangs on the latency model (fold 4, multiplication 3 and so on). That is a reasoned but not universal choice; on other hardware or with other weights the value shifts.

Where the cheapest entry passer ends.

The entry test measures to 2^31, the depth test to 2^34. The diploma (RRC-64-40, 256 streams at 1 TB) was run for the cheapest entry passer on 20–21 August 2026 — and it does NOT pass: ten streams hold the full 1 TB, two fall at **2^38** (256 GB) in the FPF test on the lowest bits, with p = 6.8×10^-19 and 3.0×10^-27. Safeguarded by a depth anchor on exactly those streams (NASAM holds 1 TB there, mix13 falls at 2^19) and by the variant using addition instead of XOR, which falls as well (2^39): **the weakness sits in the pattern, not in the leading constant.** In detail in the paper *Where K9 Ends*.

The class.

This is about hash finalizers (collisions allowed, 128-bit fold available), not about bijective generator mixers — a different class with a different cost calculation.

The nearest relatives in the literature.

Martin Leitner-Ankerl measured mixers over the same streams (RRC without complement, verified: the word does not occur in his README) with the same test program (PractRand 0.95, `-tf 2 -tlmin 10 -tlmax 40`) and plotted cost against test depth, with a declared Pareto front. Twenty mixers are listed; fifteen carry results, five are marked TODO (counted from his README, 30 August 2026). So the idea of setting those two axes against each other is not new. His cost measure is measured runtime on a particular CPU, his selection is hand-picked, and he claims no threshold. This paper starts there. It costs operations by weight rather than by wall-clock time, enumerates instead of selecting, and asks after a floor, not a ranking.

Jon Maiga systematically tabulated mixer skeletons against PractRand depth in 2020 (*Tuning bit mixers*: for PractRand-40 it takes mxmxmx or xmxmxm there — three multiplications with intermediate steps, roughly K15–K17 in our model) and later tested individual constructions with Evensen's RRC (mx3). That precedence is expressly acknowledged. The delta of this paper: a weighted cost model including the 128-bit fold, RRC disguises at every stage instead of the bare counter, enumeration instead of heuristics — a floor at K9, well below the cheapest passers there, and a diploma-capable chain at K10.

Reproducibility

The rig is documented openly: the feeder that produces the disguises, the unmodified test program (PractRand), the search. Anyone can push any chain through the ladder themselves and reproduce the verdict.

And before every run the rig proves that it measures: a known bad case (mix13) MUST fall, a known good one MUST hold. Otherwise nothing gets measured at all. A test bench that does not trip on a known bad case is broken, not thorough.

In brief

The question	How cheap can a good 64-bit hash finalizer be?
The result	On the RRC ladder nothing below cost 9 passes the entry test. K9 is the empirical floor. The diploma costs one point more: K10 passes 256 streams at 1 TB.
The reason	Two multiplicative rounds are needed; the bare K8 fold falls on counter inputs; a constant up front (+1) repairs that causally.
The counter-checks	500 constant variants (K8 fails structurally), hardware AES (K5–K8 dies at the first checkpoint, independently confirmed by Ankerl 2020), the one-round route only from K17 — and a partly enumerated K9 family (fold–bswap–fold), 1,309 of 30,980 chains decided: 23 entry passers, 0 of 23 survive the depth.
The status	Evidence + mechanism, reproducible. NOT a proof. Both bounds measured: entry at 9, diploma at 10 as an upper bound. Whether any K9 passes the diploma is open.
The value	Not a new mixer (the K9 structure is known), but the measured bounds and the named reason for them — and how little separates the two heights: one skeleton, one middle member.

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As of: 28 August 2026 (working draft, after both diploma runs). Measuring rig: *The Ladder*. The chain that failed at cost 9: *LU9* and *Where K9 Ends*. The one that holds at cost 10: *MEER10*.